

E7 Solving trigonometric equations

Name: _____

Class: _____

Date: _____

Time: **225 minutes**

Marks: **186 marks**

Comments:

Q1.

- (a) Solve the equation $\cot x = 2$, giving all values of x in the interval $0 \leq x \leq 2\pi$ in radians to two decimal places.

(2)

- (b) Show that the equation $\operatorname{cosec}^2 x = \frac{3 \cot x + 4}{2}$ can be written as

$$2 \cot^2 x - 3 \cot x - 2 = 0$$

(2)

- (c) Solve the equation $\operatorname{cosec}^2 x = \frac{3 \cot x + 4}{2}$, giving all values of x in the interval $0 \leq x \leq 2\pi$ in radians to two decimal places.

(4)

(Total 8 marks)

Q2.

- (a) (i) Express $6 \sin \theta + 8 \cos \theta$ in the form $R \sin(\theta + \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. Give your value for α to the nearest 0.1° .

(2)

- (ii) Hence solve the equation $6 \sin 2x + 8 \cos 2x = 7$, giving all solutions to the nearest 0.1° in the interval $0^\circ < x < 360^\circ$.

(4)

- (b) (i) Prove the identity $\frac{\sin 2x}{1 - \cos 2x} = \frac{1}{\tan x}$.

(4)

- (ii) Hence solve the equation

$$\frac{\sin 2x}{1 - \cos 2x} = \tan x$$

giving all solutions in the interval $0^\circ < x < 360^\circ$.

(4)

(Total 14 marks)

Q3.

- (a) Solve the equation $\sin 2x = \sin 48^\circ$, giving the values of x in the interval $0^\circ \leq x < 360^\circ$.

(4)

- (b) Solve the equation $2 \sin \theta - 3 \cos \theta = 0$ in the interval $0^\circ \leq \theta < 360^\circ$, giving your answers to the nearest 0.1° .

(4)

(Total 8 marks)

Q4.

- (a) Solve the equation $\sec x = 3$, giving the values of x in radians to two decimal places in the interval $0 \leq x < 2\pi$.

(3)

- (b) Show that the equation $\tan^2 x = 2 \sec x + 2$ can be written as $\sec^2 x - 2 \sec x - 3 = 0$.

(2)

- (c) Solve the equation $\tan^2 x = 2 \sec x + 2$, giving the values of x in radians to two decimal places in the interval $0 \leq x < 2\pi$.

(4)

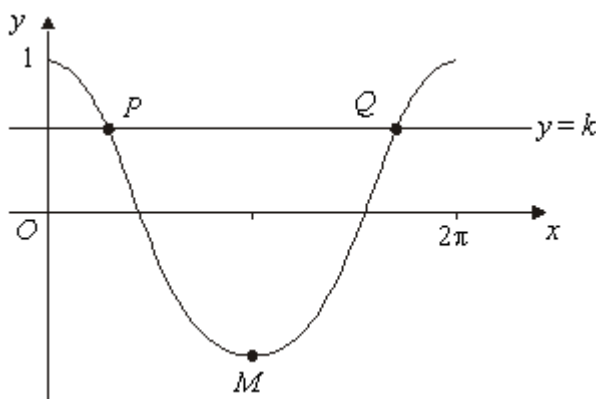
(Total 9 marks)

Q5.

- (a) Solve the equation $\cos x = 0.3$ in the interval $0 \leq x \leq 2\pi$, giving your answers in radians to three significant figures.

(3)

- (b) The diagram shows the graph of $y = \cos x$ for $0 \leq x \leq 2\pi$, and the line $y = k$.



The line $y = k$ intersects the curve $y = \cos x$, $0 \leq x \leq 2\pi$, at the points P and Q . The point M is the minimum point of the curve.

- (i) Write down the coordinates of the point M .

(2)

- (ii) The x -coordinate of P is α .

Write down the x -coordinate of Q in terms of π and α .

(1)

- (c) Describe the geometrical transformation that maps the graph of $y = \cos x$ onto the graph of $y = \cos 2x$.

(2)

- (d) Solve the equation $\cos 2x = \cos \frac{4\pi}{5}$ in the interval $0 \leq x \leq 2\pi$, giving the values of x in terms of π .

(4)

(Total 12 marks)

Q6.

- (a) (i) Show that the equation

$$2 \cot^2 x + 5 \operatorname{cosec} x = 10$$

can be written in the form $2 \operatorname{cosec}^2 x + 5 \operatorname{cosec} x - 12 = 0$.

(2)

- (ii) Hence show that $\sin x = -\frac{1}{4}$ or $\sin x = \frac{2}{3}$.

(3)

- (b) Hence, or otherwise, solve the equation

$$2 \cot^2(\theta - 0.1) + 5 \operatorname{cosec}(\theta - 0.1) = 10$$

giving all values of θ in radians to two decimal places in the interval $-\pi < \theta < \pi$.

(3)

(Total 8 marks)

Q7.

- (a) Express $\cos 2x$ in terms of $\sin x$.

(1)

- (b) (i) Hence show that $3 \sin x - \cos 2x = 2 \sin^2 x + 3 \sin x - 1$ for all values of x .

(2)

- (ii) Solve the equation $3 \sin x - \cos 2x = 1$ for $0^\circ < x < 360^\circ$.

(4)

- (c) Use your answer from part (a) to find $\int \sin^2 x \, dx$.

(2)

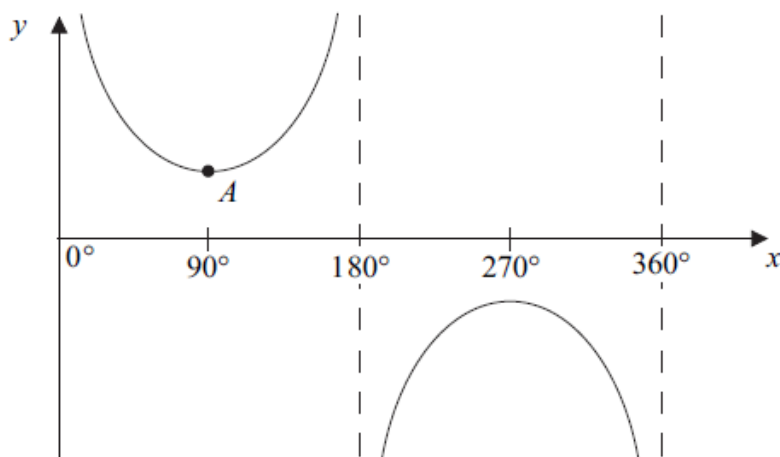
(Total 9 marks)

Q8.

- (a) Sketch the graph of $y = \tan x$ for $0^\circ \leq x \leq 360^\circ$ (3)
- (b) Write down the **two** solutions of the equation $\tan x = \tan 61^\circ$ in the interval $0^\circ \leq x \leq 360^\circ$ (2)
- (c) (i) Given that $\sin \theta + \cos \theta = 0$, show that $\tan \theta = -1$. (1)
- (ii) Hence solve the equation $\sin (x - 20^\circ) + \cos (x - 20^\circ) = 0$ in the interval $0^\circ \leq x \leq 360^\circ$ (4)
- (d) Describe the single geometrical transformation that maps the graph of $y = \tan x$ onto the graph of $y = \tan (x - 20^\circ)$. (2)
- (e) The curve $y = \tan x$ is stretched in the x -direction with scale factor $\frac{1}{4}$ to give the curve with equation $y = f(x)$. Write down an expression for $f(x)$. (1)
- (Total 13 marks)

Q9.

- (a) Solve the equation $\operatorname{cosec} x = 2$, giving all values of x in the interval $0^\circ < x < 360^\circ$. (2)
- (b) The diagram shows the graph of $y = \operatorname{cosec} x$ for $0^\circ < x < 360^\circ$.



Sketch the graph of $y = |\operatorname{cosec} x|$ for $0^\circ < x < 360^\circ$.

(2)
(Total 4 marks)

Q10.

- (a) Express $4 \cos x + 3 \sin x$ in the form $R \cos (x - \alpha)$, where $R > 0$ and $0^\circ < \alpha < 360^\circ$, giving your value for α to the nearest 0.1° . (3)
- (b) Hence solve the equation $4 \cos x + 3 \sin x = 2$ in the interval $0^\circ < x < 360^\circ$, giving all solutions to the nearest 0.1° . (4)
- (c) Write down the minimum value of $4 \cos x + 3 \sin x$ and find the value of x in the interval $0^\circ < x < 360^\circ$ at which this minimum value occurs. (3)
- (Total 10 marks)**

Q11.

- (a) Describe the geometrical transformation that maps the curve with equation $y = \sin x$ onto the curve with equation:
- (i) $y = 2 \sin x$; (2)
- (ii) $y = -\sin x$; (2)
- (iii) $y = \sin(x - 30^\circ)$. (2)
- (b) Solve the equation $\sin(\theta - 30^\circ) = 0.7$, giving your answers to the nearest 0.1° in the interval $0^\circ \leq \theta \leq 360^\circ$. (3)
- (c) Prove that $(\cos x + \sin x)^2 + (\cos x - \sin x)^2 = 2$. (4)
- (Total 13 marks)**

Q12.

It is given that $2 \operatorname{cosec}^2 x = 5 - 5 \cot x$.

- (a) Show that the equation $2 \operatorname{cosec}^2 x = 5 - 5 \cot x$ can be written in the form

$$2 \cot^2 x + 5 \cot x - 3 = 0$$

(2)

- (b) Hence show that $\tan x = 2$ or $\tan x = -\frac{1}{3}$. (2)

- (c) Hence, or otherwise, solve the equation $2\operatorname{cosec}^2 x = 5 - 5 \cot x$, giving all values of x in radians to one decimal place in the interval $-\pi < x \leq \pi$.

(3)

(Total 7 marks)

Q13.

It is given that $3 \cos \theta - 2 \sin \theta = R \cos(\theta + \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$.

- (a) Find the value of R .

(1)

- (b) Show that $\alpha \approx 33.7^\circ$.

(2)

- (c) Hence write down the maximum value of $3 \cos \theta - 2 \sin \theta$ and find a **positive** value of θ at which this maximum value occurs.

(3)

(Total 6 marks)

Q14.

- (a) Describe the single geometrical transformation by which the curve with equation $y = \tan \frac{1}{2}x$ can be obtained from the curve $y = \tan x$.

(2)

- (b) Solve the equation $\tan \frac{1}{2}x = 3$ in the interval $0 < x < 4\pi$, giving your answers in radians to three significant figures.

(4)

- (c) Solve the equation

$$\cos \theta (\sin \theta - 3 \cos \theta) = 0$$

in the interval $0 < \theta < 2\pi$, giving your answers in radians to three significant figures.

(5)

(Total 11 marks)

Q15.

- (a) Solve the equation $\sec x = 5$, giving all the values of x in the interval $0 \leq x \leq 2\pi$ in radians to two decimal places.

(3)

- (b) Show that the equation $\tan^2 x = 3 \sec x + 9$ can be written as

$$\sec x^2 - 3 \sec x - 10 = 0$$

(2)

- (c) Solve the equation $\tan^2 x = 3 \sec x + 9$, giving all the values of x in the interval $0 \leq x \leq 2\pi$ in radians to two decimal places.

(4)

(Total 9 marks)

Q16.

- (a) (i) Express $\sin 2x$ in terms of $\sin x$ and $\cos x$.

(1)

- (ii) Express $\cos 2x$ in terms of $\cos x$.

(1)

- (b) Show that

$$\sin 2x - \tan x = \tan x \cos 2x$$

for all values of x .

(3)

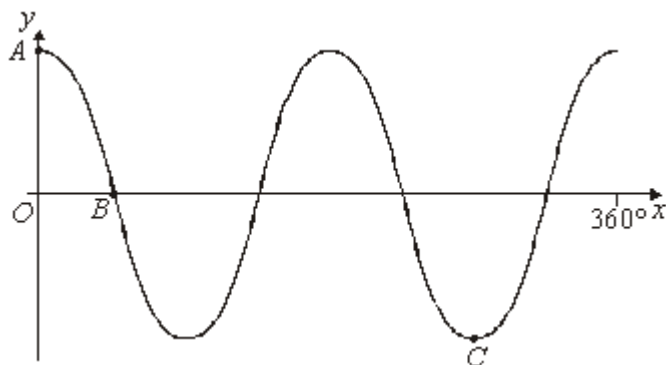
- (c) Solve the equation $\sin 2x - \tan x = 0$, giving all solutions in degrees in the interval $0^\circ < x < 360^\circ$.

(4)

(Total 9 marks)

Q17.

The diagram shows the graph of $y = \cos 2x$ for $0^\circ \leq x \leq 360^\circ$.



- (a) Write down the coordinates of the points A , B and C marked on the diagram.

(4)

- (b) Describe the single geometrical transformation by which the curve with equation $y = \cos 2x$ can be obtained from the curve with equation $y = \cos x$.

(2)

- (c) Solve the equation

$$\cos 2x = 0.37$$

giving all solutions to the nearest 0.1° in the interval $0^\circ \leq x \leq 360^\circ$.
 (No credit will be given for simply reading values from a graph.)

(5)

(Total 11 marks)

Q18.

- (a) (i) Show that the equation

$$4 \tan \theta \sin \theta = 15$$

can be written as

$$4 \sin^2 \theta = 15 \cos \theta$$

(1)

- (ii) Use an appropriate identity to show that the equation

$$4 \sin^2 \theta = 15 \cos \theta$$

can be written as

$$4 \cos^2 \theta + 15 \cos \theta - 4 = 0$$

(2)

- (b) (i) Solve the equation $4c^2 + 15c - 4 = 0$.

(2)

- (ii) Hence explain why the only value of $\cos \theta$ which satisfies the equation

$$4 \cos^2 \theta + 15 \cos \theta - 4 = 0$$

$$\text{is } \cos \theta = \frac{1}{4}.$$

(1)

- (iii) Hence solve the equation $4 \tan \theta \sin \theta = 15$ giving all solutions to the nearest 0.1° in the interval $0^\circ \leq \theta \leq 360^\circ$.

(2)

- (c) **Write down** all the values of x in the interval $0^\circ \leq x \leq 90^\circ$ for which

$$4 \tan 4x \sin 4x = 15$$

giving your answers to the nearest degree.

(2)

(Total 10 marks)

Q19.

It is given that $\tan^2 x = \sec x + 11$.

- (a) Show that the equation $\tan^2 x = \sec x + 11$ can be written in the form

$$\sec^2 x - \sec x - 12 = 0$$

(2)

- (b) Hence show that $\cos x = \frac{1}{4}$ or $\cos x = -\frac{1}{3}$.

(3)

- (c) Hence, or otherwise, solve the equation $\tan^2 x = \sec x + 11$, giving all values of x to the nearest degree in the interval $0^\circ < x < 360^\circ$.

(3)

(Total 8 marks)

Q20.

- (a) Express $2 \sin x + \cos x$ in the form $R \sin(x + \alpha)$ where R is a positive constant and α is an acute angle. Give your value of α to the nearest 0.1° .

(3)

- (b) Solve the equation $2 \sin x + \cos x = 1$ for $0^\circ \leq x < 360^\circ$.

(4)

(Total 7 marks)

Mark schemes

Q1.

(a) $\cot x = 2 \Rightarrow \tan x = 0.5$

M1

$x = 0.46, 3.61$

AWRT; no others within range

A1

2

(b) $\operatorname{cosec}^2 x = \frac{3 \cos x + 4}{2}$

$2(1 + \cot^2 x) = 3 \cot x + 4$

Correct use of $\operatorname{cosec}^2 x = 1 + \cot^2 x$

M1

$(2 \cot^2 x - 3 \cot x + 2 - 4 = 0)$

$2 \cot^2 x - 3 \cot x - 2 = 0$

*AG; correct with no slips from line
with no fractions*

A1

2

(c) $(2 \cot x + 1)(\cot x - 2) (= 0)$

Attempt to solve

M1

$\cot x = -\frac{1}{2}, 2$

A1

$\tan x = -2, 0.5$

$x = 0.46, 3.61, 2.03, 5.18$

2 correct Allow 3.6(0)

B1

*4 correct (with no extras in range) AWRT
SC Degrees*

$$\left. \begin{array}{l} 26.57, 206.57 \\ 116.57, 296.57 \end{array} \right\} \text{ B1 For 2 correct}$$

B1

4

[8]

Q2.

(a) (i) $R = 10$

$$R = 10$$

B1

$$\tan \alpha = \frac{8}{6}, \quad \alpha = 53.1$$

For α ; ft incorrect R

B1F

2

(ii) $\sin (2x + 53.1) = 0.7$

M1

$$2x + 53.1 = 44.4$$

one correct answer ; ft α and R

A1F

$$135.6 \text{ or } 135.7, 404.4, 495.6 \text{ or } 495.7$$

3 other correct answers – ignore extras

A1

$$x = 41.2 \text{ or } 41.3, 175.6 \text{ or } 175.7, \\ 221.2 \text{ or } 221.3, 355.6 \text{ or } 355.7$$

four solutions

CAO (with decimal place discrepancies)

Answers only: 0/4

A1

4

(b) (i) $\sin 2x = 2 \sin x \cos x$

B1

$$\cos 2x = \cos^2 x - \sin^2 x$$

identities for $\sin 2x$ and $\cos 2x$ in any correct form

B1

$$\frac{\sin 2x}{1 - \cos 2x} = \frac{2 \sin x \cos x}{1 - (1 - 2 \sin^2 x)} =$$

use of candidate's double angle formulae

M1

$$\frac{2 \sin x \cos x}{2 \sin^2 x} = \frac{\cos x}{\sin x} = \frac{1}{\tan x}$$

AG, CSO

A1

4

(ii) $\frac{1}{\tan x} = \tan x \quad \tan x = \pm 1$
*(see * below)*

M1A1

$x = 45,$

B1

135, 225, 315

if answers given without working, B1 max

*$\frac{1}{\tan x} = \tan x$ seen and followed by
 correct answers without working 4 out of 4*

A1

4

** If hence ignored, so working in sines and
 cosines, must simplify as far as:*

$\cos^2 x = \sin^2 x \quad \text{or} \quad \cos^2 x = \frac{1}{2} \quad \text{or} \quad \sin^2 x = \frac{1}{2} \quad \text{for M1}$				
$\cos 2x = 0 \quad \text{or} \quad \cos x = \pm \frac{1}{\sqrt{2}} \quad \text{or} \quad \sin x = \pm \frac{1}{\sqrt{2}} \quad \text{for A1}$				

[14]

Q3.

(a) $2x = 48$

PI by $x = 24^\circ$

B1

$$2x = 180 - 48$$

Accept equivalents for x

M1

$$2x = 360 + 48 \text{ and } 2x = 360 + 180 - 48$$

Accept equivalents for x

M1

$$x = 24^\circ, 66^\circ, 204^\circ, 246^\circ$$

CAO; need all four, no extras in given interval

A1

4

(b) $\frac{\sin \theta}{\cos \theta} = \tan \theta$

Stated or used

M1

$$2\sin \theta - 3 \cos \theta = 0 \Rightarrow \tan \theta = 1.5$$

A1

$$\theta = 56.3^\circ$$

Condone > 1dp

A1

$$\theta = 56.3^\circ + 180^\circ = 236.3^\circ$$

*Ft on c's PV + 180° dep only on the M1
provided no 'extra' solutions in the given
interval.*

A1F

4

[8]

Q4.

(a) $x = \cos^{-1} \frac{1}{3}$

PI

M1

$$= 1.23, 5.05 \text{ (} 0.39\pi, 1.61\pi \text{)}$$

*AWRT (−1 for each error in range)
SC 70.53, 289.47 B1*

A1,A1

3

(b) $\sec^2 x - 1 = 2 \sec x + 2$
use of $\sec^2 x = 1 + \tan^2 x$

M1

$\sec^2 x - 2 \sec x - 3 = 0$
AG; CSO

A1

2

(c) $\sec^2 x - 2 \sec x - 3 = 0$
 $(\sec x - 3)(\sec x + 1) = 0$
attempt to solve

M1

$\cos x = \frac{1}{3}$ or -1 o.e

A1

$x = 1.23, 5.05,$
(2 answers in range from (a)) AWRT

B1F

3.14 (π)
all correct and no extras in range
SC 70.53, 289.47, 180 B1

B1

4

[9]

Q5.

(a) $\{x =\} \cos^{-1} (0.3) = 1.266.... \{ = \beta \}$
 $\cos^{-1} (0.3)$ PI by eg 72° or 73°

M1

$\{x =\} 2\pi - \beta$
Condone degrees or mix.

m1

$x = 1.27, 5.02$

Accept 1.26 to 1.27 with 5.01 to 5.02 inclusive

		A1	3
(b)	(i) $m(\pi, -1)$ <i>B1 for each coordinate</i>		
		B1;B1	2
	(ii) $\{x_Q = \} = 2\pi - \alpha$ <i>OE (unsimplified)</i>		
		B1	1
(c)	Stretch (I) in x -direction (II) scale <i>Need(I) & one of (II),(III)</i>		
		M1	
	factor $\frac{1}{2}$ (III)		
		A1	2
(d)	$\cos 2x = \cos \frac{4\pi}{5} \Rightarrow 2x = \frac{4\pi}{5}$ <i>OE. (From correct work)</i>		
		B1	
	$\Rightarrow x = \frac{2\pi}{5} (= \alpha)$ <i>Condone decimals/degrees</i>		
	$x = \pi - \alpha$; OE		
	OE eg $2x = 2\pi - \frac{4\pi}{5}$ <i>Correct quadrant; condone degrees/decimals/mix</i>		
		M1	
	$x = \pi + \alpha$; $x = 2\pi - \alpha$; OE <i>Need both (OE for $2x =$) with no extras (quadrants) within the given interval. Condone degrees/decimals/mix</i>		

m1

$$x = \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{7\pi}{5}, \frac{8\pi}{5}$$

*Need all 4 solutions for x but condone unsimplified provided in terms of π
Ignore extra values outside the given interval.*

A1

4

[12]

Q6.

(a) (i) $2 (\operatorname{cosec}^2 x - 1) + 5 \operatorname{cosec} x = 10$

M1

$$2 \operatorname{cosec}^2 x - 2 + 5 \operatorname{cosec} x - 10 = 0$$

$$2 \operatorname{cosec}^2 x + 5 \operatorname{cosec} x - 12 = 0$$

AG

A1

2

(ii) $(2 \operatorname{cosec} x - 3)(\operatorname{cosec} x + 4) = 0$

Attempt to solve

M1

$$\operatorname{cosec} x = \frac{3}{2} \text{ or } -4$$

Condone answers with no method shown

A1

$$\sin x = \frac{2}{3} \text{ or } -\frac{1}{4}$$

AG

A1

3

(b) $(\theta - 0.1) = 0.73, 2.41, -0.25, -2.89$ AWRT

*2 correct values, may be implied later
(41.8, 138.2, -165.5, -14.5)*

B1

$\theta = 0.83, 2.51, -0.15, -2.79$ AWRT
 2 correct answers

B1

+ 2 correct answers and no extra within range

B1

3

[8]

Q7.

(a) $\cos 2x = 1 - 2\sin^2 x$

B1

1

(b) (i) $3\sin x - \cos 2x = 3\sin x - (1 - 2\sin^2 x)$
 Candidate's $\cos 2x$ or $\sin^2 x$

M1

$= 3\sin x - 1 + 2\sin^2 x$
 AG

A1

2

(ii) $2\sin^2 x + 3\sin x - 2 = 0$
 Soluble quadratic form

M1

$(2\sin x - 1)(\sin x + 2) = 0$
 Attempt to solve (allow one error in formula,
 allow sign errors)

M1

$\sin x = \frac{1}{2} \quad x = 30 \quad x = 150$

M1

$\sin^{-1}$ and two solutions ($0^\circ < x < 360^\circ$)
 A0 if radians

A1

4

Allow misread for $2\sin^2 x + 3\sin x - 1 = 0$
 Soluble quadratic form

(M1)

$$\sin x = \frac{-3 \pm \sqrt{17}}{4}$$

Use of formula (allow one error)

(M1)

$$x = 16.3^\circ, 163.7^\circ$$

Max 3/4

(A1)

(c) $\int \frac{1}{2}(1 - \cos 2x) = \frac{x}{2} - \frac{\sin 2x}{4} (+c)$

*M1 – solve integral, must have 2 terms
for $\sin^2 x$ from (a)*

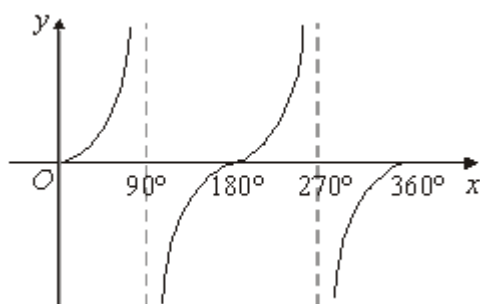
M1A1

2

[9]

Q8.

(a)



*Correct shape of branch from O {to 90°}
or correct shapes of branches from 90°- 360°*

M1

*Complete graph for $0^\circ \leq x \leq 360^\circ$
(Asymptotes not explicitly required but graphs
should show "tendency")*

A1

Correct scaling on x-axis $0^\circ \leq x \leq 360^\circ$

A1

3

(b) 61° ;

For 61°

B1

241°

For 241° and no 'extras' in the interval $0^\circ \leq x \leq 360^\circ$

B1

2

(c) (i) $\sin \theta = -\cos \theta \Rightarrow \frac{\sin \theta}{\cos \theta} = -1$

$\Rightarrow \tan \theta = -1$

AG; be convinced that the identity $\frac{\sin \theta}{\cos \theta} = \tan \theta$ is known and validly used

B1

1

(ii) $\Rightarrow \tan(x - 20^\circ) = -1$

M1

$x - 20^\circ = \tan^{-1}(-1)$

m1

$x - 20^\circ = 135^\circ, 315^\circ \dots$

$x = 155^\circ;$

A1

335°

Ft on (180 + "155") and no 'extras' in the given interval.

A1ft

4

(d) Translation

'Translation'/'translate(d)'

B1

$$\begin{bmatrix} 20 \\ 0 \end{bmatrix}$$

Accept equivalent in words provided linked to 'translation/move/shift' (Note: B0B1 is possible)

B1

2

(e) $f(x) = \tan 4x$

For $\tan 4x$

B1

1

[13]

Q9.

(a) $\operatorname{cosec} x = 2$

$$\Rightarrow \sin x = \frac{1}{2}$$

30° scores M1 implied

M1

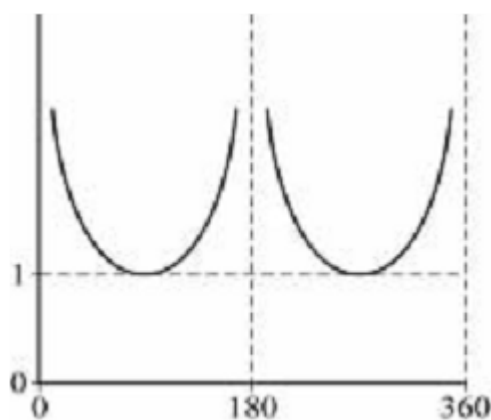
$= 30, 150$

and no extras in range

A1

2

(b)



all positive, 2 U shapes

M1

*minima consistent > 0 , **not** intersecting with each other or y-axis*

A1

2

[4]

Q10.

(a) $R = 5$

B1

$$\tan \alpha = \frac{3}{4} \text{ (OE)} \quad \alpha = 36.9^\circ \text{ (ISW 216.9)}$$

$$\text{SC1 } \tan \alpha = \frac{4}{3}, \alpha = 53.1^\circ$$

R, \alpha, PI in (b)

M1A1

3

(b) $\cos(x - \alpha) = \frac{2}{R}$

M1

$$x - \alpha = 66.4^\circ$$

A1

$$x = 103.3^\circ$$

A1F

$$x = 330.4^\circ$$

*accept 330.5°, -1 each extra
ft on acute \alpha*

A1F

4

(c) minimum value = -5
ft on R

B1F

$$\cos(x - 36.9) = -1$$

SC cos (x + 36.9) treat as miscopy

M1

$$x = 216.9^\circ$$

*216.9 or better accept graphics calculator
solution to this accuracy*

SC Find max:

max = 5 at (x + 36.9) + stated 1/3

Max 8/10 for work in radians

A1

3

Q11.

- (a) (i) Stretch **(I)** in y -direction **(II)**

Scale factor 2 **(III)**

>1 transformation is M0.

M1 for **(I)** and either **(II)** or **(III)** or **(III)**

M1A1

2

- (ii) Reflection;

M1

in x -axis

'Reflection'/'reflect(ed)'

(or in y -axis or $y = 0$ or $x = 0$)

A1

2

- (iii) Translation;

'Translation'/'translate (d)'

B1

$$\begin{bmatrix} 30^\circ \\ 0 \end{bmatrix}$$

Accept **full** equivalent in words provided linked to 'translation/move/shift' and **positive** x -direction
(Note: B0 B1 is possible)

B1

2

- (b) $\{\theta - 30^\circ = \sin^{-1}(0.7) = 44.4\dots^\circ$

Inverse sine of 0.7 PI eg by sight of 44, 74 or better

M1

$$\dots\dots\dots = 180^\circ - 44.4^\circ$$

Valid method for 2nd angle

m1

$$\theta = 74.4^\circ, 165.6^\circ$$

Condone >1 dp accuracy

A1

(c) $\dots = \cos^2 x + 2\cos x \sin x + \sin^2 x + \cos^2 x - 2\cos x \sin x + \sin^2 x$
Award for either bracket expanded correctly

M1

$$\dots = 2\cos^2 x + 2\sin^2 x$$

OE

A1

$$= 2(\cos^2 x + \sin^2 x) = 2 (1)$$

$\cos^2 x + \sin^2 x = 1$ stated or used.

M1

$$= 2$$

AG (be convinced)

A1

4

[13]

Q12.

(a) $2\operatorname{cosec}^2 x = 5(1 - \cot x)$

$$2 + 2 \cot^2 x = 5 - 5 \cot x$$

use of $\operatorname{cosec}^2 x = 1 + \cot^2 x$

M1

$$2 \cot^2 x + 5 \cot x - 3 = 0$$

AG

A1

2

(b) $(2 \cot x - 1)(\cot x + 3) = 0$
 or $2 + 5t - 3t^2 = 0$ Or in $\tan x$
 $(2 - t)(1 + 3t) = 0$

M1

$$\cot x = \frac{1}{2}, -3$$

$$\tan x = 2, -\frac{1}{3}$$

AG

A1

2

(c) $x = 1.1, -2.0$ AWRT

$x = -0.3, 2.8$ AWRT

Any 2 correct (In degrees): B0

B1

Any 3 correct (In degrees): B1

B1

4 correct (In degrees): B2

B1

[7]

Q13.

(a) $R = \sqrt{13}$ Or 3.6

B1

1

(b) $\frac{\sin \alpha}{\cos \alpha} = \tan \alpha = \frac{2}{3}$ $\alpha \approx 33.7$

Allow M1 for $\tan \alpha = \frac{-2}{3}$ or $\pm \frac{3}{2}$
AG convincingly obtained

M1A1

2

(c) maximum value = $\sqrt{13}$

B1F

$\cos(\theta + 33.7) = 1$ ($\theta = -33.7$)

M1

$\theta = 326.3$

AWRT 326

A1

3

[6]

Q14.

- (a) Stretch (I) in x -direction (II) scale
Need(I) and one of (II),(III)

M1

factor 2 (III)

M0 if more than one transformation

A1

2

- (b) $\tan^{-1} 3 = 1.2(49....) (= \alpha)$
 $\tan^{-1} 3$ [PI by 71.(56..)]

M1

$$\left\{ \frac{1}{2}x \right\} = \pi + \alpha;$$

Correct quadrant; condone degrees or mix

m1

$$\frac{1}{2}x = 1.249...; 4.3906...$$

$$x = 2.498... = 2.50 \text{ to } 3 \text{ sf}$$

A1

$$x = 8.781... = 8.78 \text{ to } 3 \text{ sf}$$

A1

*Condone 2.5 otherwise deduct **max** of 1 mark throughout this question from A marks if 'correct' rads. but to 2sf or final answers in degrees. (143°, 503°)*

As usual, accept greater accuracy answers. Ignore extra values outside the given interval (0 to 12.6). If > 2 values inside interval lose an A mark for each one.

NB M1m0A1A0 is possible

SC after M0 for error $\tan x = 6$;

Either $x = 1.40(5)$, $4.54(7)$, $7.68(8)$, $10.8(3)$ or $x = 80.5^\circ$, 260.5° , 440.5° , 620.5° SC
 B1

(accept each rounded or truncated to 3 sf)

4

- (c) $\cos \theta = 0$, $\sin \theta - 3 \cos \theta = 0$
Need both

M1

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \text{ or } \tan \theta = 3$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \text{ seen/used}$$

M1

$$\cos \theta = 0 \Rightarrow \theta = \frac{\pi}{2} = 1.57(07\dots)$$

Accept $\frac{\pi}{2}$

B1

$$\text{or } \theta = \frac{3\pi}{2} = 4.71(23\dots)$$

Accept $\frac{3\pi}{2}$

B1

$$\tan \theta = 3 \Rightarrow \theta = 1.249\dots; 4.3906\dots = 1.25, 4.39 \text{ to 3sf}$$

If not correct, ft on (b)

NB M0M1(B0B0)A1ft is possible

A1 ✓

$$90^\circ; 270^\circ; 71.5(6)^\circ; 251.5(6)^\circ$$

5

[11]

Q15.

(a) $\sec x = 5$
 $\cos x = 0.2$

M1

$$x = 1.37, 4.91 \text{ AWRT}$$

A1A1

3

(b) $\tan^2 x = 3 \sec x + 9$
 $\sec^2 x - 1 = 3 \sec x + 9$

for using $\sec^2 x = 1 + \tan^2 x$ OE

M1

$$\sec^2 x - 3 \sec x - 10 = 0$$

AG

A1

2

- (c) $(\sec x - 5)(\sec x + 2) = 0$
or use of formula (attempt)

M1

$$\sec x = 5, -2$$

A1

$$\cos x = 0.2, -0.5$$

$$x = 1.37, 4.91$$

any 2 correct or ft their 2 answers in (a)

B1F

$$2.09, 4.19$$

all 4 correct, no extras

A1

4

[9]

Q16.

(a) (i) $\sin 2x = 2 \sin x \cos x$

B1

1

(ii) $\cos 2x = 2 \cos^2 x - 1$

B1

1

(b) $\sin 2x - \tan x = 2 \sin x \cos x - \frac{\sin x}{\cos x}$

Use of their $\cos 2x$ or $\sin 2x$

M1

$$= \sin x \left(2 \cos x - \frac{1}{\cos x} \right)$$

$\frac{\sin x}{\cos x}$
 Use of $\tan x = \frac{\sin x}{\cos x}$ and the other double angle identity

M1

$$= \sin x \left(\frac{2 \cos^2 x - 1}{\cos x} \right) = \tan x \cos 2x$$

AG convincingly obtained

A1

3

(c) $\tan x \cos 2x = 0 \quad x = 180$

Ignore $x = 0$, $x = 360^\circ$ & any others outside range

B1

$$\cos 2x = 0 \text{ or } \cos^2 x = \frac{1}{2} \left(\text{or } \sin^2 x = \frac{1}{2} \right)$$

M1

$$x = 45$$

A1

$$x = 135, 225, 315$$

CAO max 3/4 for answers in radians

B1

4

[9]

Q17.

(a) $A(0^\circ, 1)$

Condone radians

B1

$$B(45^\circ, 0)$$

Condone (0.785, 0) or better.

B1

$$C(270^\circ, -1)$$

B1 for 270; B1 for -1

B1, B1

- (b) Stretch (I) in x -direction (II) with a scale factor $\frac{1}{2}$ (III)
More than one transformation is M0
*M1 for (I) **and** either (II) or (III)*

M1A1

2

- (c) $\cos^{-1}0.37 = "68.284..." (= \alpha)$
Cos⁻¹0.37 (PI eg by 68.3 or 1.19)

M1

$$x = \frac{\alpha}{2} = 34.1(42.)^\circ$$

Condone 34.2°, 34° or 0.596 rads

A1

$$x = 180 - \frac{\alpha}{2}$$

OE eg $2x = 360 - \alpha$

m1

$$x = 180 + \frac{\alpha}{2} \quad \text{and} \quad x = 180 + 180 - \frac{\alpha}{2}$$

*OE Need both (OE for $2x =$)
 with no extras (quadrants)
 within the given interval*

m1

$$2x = 38.284...; 291.175...;$$

$$428.284...; 651.715...$$

$$x = (34.1^\circ;)$$

$$145.9^\circ; 214.1^\circ; 325.9^\circ$$

*Dep. on **all** three method marks. Must be in degrees*

A1

5

[11]

Q18.

(a) (i) $4 \frac{\sin \theta}{\cos \theta} \sin \theta = 15$

$\Rightarrow 4 \sin^2 \theta = 15 \cos \theta$

AG Be convinced

B1

1

(ii) $\sin^2 \theta + \cos^2 \theta = 1$

OE seen

M1

$4(1 - \cos^2 \theta) = 15 \cos \theta$

$4 \cos^2 \theta + 15 \cos \theta - 4 = 0$

AG Be convinced

A1

2

(b) (i) $(4c - 1)(c + 4) = 0$

Factorisation or formula or completion of square

M1

$c = -4, c = \frac{1}{4}$

Both values

A1

2

(ii) Since $-1 \leq \cos \theta \leq 1$, the only possible value for $\cos \theta$ is $\frac{1}{4}$

AG convincingly explained

(Condone strict inequalities)

ft provided candidates answers for c

are $\frac{1}{4}$ and a value

k such that $k > 1$ or $k < -1$

E1ft

1

(iii) $\theta = 75.5^\circ$

B1

$\theta = 284.5^\circ$

Ft on $[360 - c's 75.5^\circ]$ as only other solution in the given interval

B1ft
2

(c) $\Rightarrow \cos 4x = \frac{1}{4}$

Links with previous parts. P1

M1

$x = 19^\circ, 71^\circ$

ft on (iii)/4....(only ft if 2 answers in given range).

A1ft
2

[10]

Q19.

(a) $\tan^2 x = \sec x + 11$

Or attempt to form quadratic in $\cos^2$

$\sec^2 x - 1 = \sec x + 11$

$\tan^2 x = \sec^2 x - 1$

M1

$\sec^2 x - \sec x - 12 = 0$

AG

A1
2

(b) $(\sec x - 4)(\sec x + 3) = 0$

attempt at solving quadratic

M1

$\sec x = 4, -3$

A1F

$\therefore \cos x = \frac{1}{4}, -\frac{1}{3}$

AG; (A0 if no use of $\cos x = \frac{1}{\sec x}$)

A1
3

(c) $x = 76^\circ, 284^\circ$

2 correct

B1

$x = 109^\circ, 251^\circ$ (or better)

other answers (–1 each extra in range)

B1,B1

3

If radians $x = 1.32, 4.97$

$1.91, 4.37$

B1 any 2 correct

B1 other 2 correct

[8]

Q20.

(a) $R = \sqrt{5}$ (or $\sqrt{1+2^2}$ or 2.23 or 2.24)

B1

$$\frac{\sin \alpha}{\cos \alpha} = \frac{1}{2} \quad \alpha = 26.6^\circ$$

CAO

SC 63.4° 1/2

M1A1

3

(b) $\sin (x + 26.6) = \frac{1}{\sqrt{5}}$

M1

$$x = \sin^{-1} \left(\frac{1}{\sqrt{5}} \right) - 26.6$$

B1 $x = 0^\circ$

m1

$x = 0^\circ$ $x = 126.8^\circ$ (126.9°, or 127 or 126° with working)

lose if extra solution in range

SC calculator trace: 126.9 full marks

B1

SC ft from 63.4°



A1

4

[7]

Examiner reports

Q1.

Part (a) was well answered by the majority of candidates. $\tan x = 0.5$ and $x = 0.46, 3.60$ was a common error. The second angle was often incorrect. Few cases of angles in degrees were seen.

In part (b), most candidates used the correct identity and were successful in answering this part of the question. The main error was $\operatorname{cosec}^2 x = \cot^2 x - 1$ followed by fudging of the rest of the solution.

In part (c), most candidates attempted to factorise the quadratic expression, although some used the quadratic formula. Those who factorised were usually correct, although solutions of $\frac{1}{2}$ and -2 were not uncommon. Those who used the formula often made the error of $\cot^2 x = 2$ or $\cot^2 x = -0.5$. Candidates with 3.60 as a solution in (a) were able to recover here but often failed to obtain both marks; 5.17 was a common error.

Q2.

There was a full range of marks seen in response to this question. Some candidates were able to complete the whole question successfully, including finding all four solutions to the particularly demanding part (a)(ii). Other candidates just scored 1 or 2 marks, from part (a)(i) and/or a correct identity in part (b)(i). Most candidates knew what to do for part (a)(i)

and got the value of R correct, although a few got $\sqrt{80}$. Many, however, had $\tan \alpha = \frac{6}{8}$. Most also knew that they were to use the expression from part (a)(i) in part (a)(ii), although a few attempted to manipulate the given expression with double angle formulae and usually abandoned the attempt. Of those using the expected approach, most scored 2 marks, some getting a third solution as well but not many finding all four. An error which was sometimes seen was to replace $2x + 53.1$ with $\theta + 106.2$.

Part (b) was generally more successful, with most candidates scoring a mark for at least one of the identities for $\sin 2x$ or $\cos 2x$ correct and many going on to use these convincingly in obtaining the requested result. Others made errors in signs and coefficients and fudged the result rather than looking for their errors. Some candidates, however, did seek an error and attempted to overwrite their work rather than starting again, often making it difficult to read what they had actually intended as their answer.

Most candidates who attempted part (b)(ii) scored at least 2 marks — the starting equation and the solution $x = 45$ — but many also obtained all four solutions. Those candidates who ignored the hence and again expanded the left-hand side were largely unable to complete the question successfully, although some did succeed via this route.

Q3.

This question, on solving trigonometrical equations, was the worse answered question on the paper. In part (a), although the majority of candidates scored the mark for writing $2x = 48$ (or $x = 24$) many failed to score any of the remaining three marks because they just quoted 156 (from $x = 180 - 24$) as the only other solution.

In part (b) candidates generally quoted the correct identity, $\tan \theta = \frac{\sin \theta}{\cos \theta}$, but a significant number of these candidates were unable to use it to derive the correct value for $\tan \theta$. Those other candidates who tried to solve the equation by initially squaring both sides of the given equation were generally less successful.

Q4.

In general this question was done well by candidates of all abilities. Part (a) was very well answered by the majority of candidates, although $\tan x = 1/3$ and $\sin x = 1/3$ were also seen. 1.23 was usually seen but the second result was often incorrect. There were very few cases of results given in degrees.

Most candidates used the correct identity part (b) in and were successful in answering this part of the question. The main error was candidates using $\tan^2 x = \sec^2 x + 1$ and then fudging the rest of the solution.

In part (c) most candidates attempted to factorise the result from part (b), although some used the quadratic formula. Those who factorised were usually correct, although solutions of $-1/3$ and 1 were not uncommon. For the final 3 answers many totally correct solutions were seen. Candidates with an incorrect solution in part (a) were able to recover here from follow through marks, but often candidates failed to obtain both marks since 0 or 6.18 often accompanied 3.14 .

Q5.

Most candidates were able to give a correct value in part (a) but a significant minority did not give any second value, despite the hint in the question: “answers”. A common wrong answer was “4.407”. In part (b)(i), some candidates mixed up the coordinates and in part (b)(ii) “ $(\pi + a)$ ” was the usual wrong answer. The description of the required transformation in part (c) was answered better this year.

Although some excellent solutions were seen to part (d), these were very much the exception. The vast majority of candidates could not deal correctly with the $2x$ beyond

finding the first solution, $x = \frac{2\pi}{5}$.

Q6.

Parts (a) and (b) were very well answered with most candidates scoring full marks. Surprisingly, part (c) proved to be demanding. Many candidates only found 2 rather than 4 solutions and, more alarmingly, candidates seemed to have a problem dealing with the $(\theta - 0.1)$ and often subtracted the 0.1, giving an answer of 0.63 rather than 0.83.

Q7.

Most candidates just wrote down the required expression for $\cos 2x$, although some worked it out from $\cos^2 x - \sin^2 x$ or $\cos(A + B)$. Omission of the 2 was a relatively rare error. Most candidates were able to derive the result in part (b)(i) although some were penalised for being too brief. Most candidates too were able to solve the resulting quadratic equation, with some making it a little more difficult than intended by reading the

1 on the right hand side as 0. Most candidates found two results for angle x in the required range. Part (c) was less successful: although most candidates understood that they needed the integral in terms of $\cos^2 x$, there were many algebraic slips, and sign and coefficient errors in the integral of $\cos^2 x$ itself.

Q8.

Candidates usually have difficulty with trigonometry and this session was no exception. Sketching of the graph of $y = \tan x$ was generally not done well and perhaps illustrated the over-reliance on calculators for graphical work. Surprisingly the first branch, between $x = 0^\circ$ and $x = 90^\circ$, was missing in a significant number of sketches. In part (b) candidates were expected to be able to just write down the two values by first noting by inspection that $x = 61$ was a solution. Instead many gave the incorrect solutions $\tan 61$ and $\tan 241$.

Candidates' solutions to show that $\tan \theta = -1$ in part (c)(i) were often unconvincing. The examiners were generally looking for a division by $\cos \theta$ before or after a rearrangement.

Part (c)(ii) starts with the word 'Hence' so those candidates who only gave the answers 155° and 335° gained no credit. Those candidates who spotted the link and indicated ' $\tan(x - 20^\circ) = -1$ ' generally went on to gain full marks, although some others subtracted (instead of added) 20 from their answers to part (c)(i). A common error was to start by writing ' $\sin(x - 20) + \cos(x - 20) = \sin x - 20 \sin + \cos x - \cos 20$ ' from which candidates never recovered even after a page of working.

Many candidates gained at least 1 mark for their description of the geometrical transformation in part (d). The final part of the question was answered well by the above average candidates but some left their answer as $f(4x)$ which was not awarded the mark. A substantial number of candidates gave the answer as $4f(x)$.

Q9.

Part (a) was answered very well by the majority of candidates. Where marks were lost, it was generally for an incorrect or omitted second angle in the interval.

Part (b) was also very well answered, with few incorrect responses.

Q10.

Part (a) was done well by most candidates, although some did a lot of unnecessary work in obtaining the correct answer. It should be possible to write down the values of R and $\tan \alpha$ from the expression.

For part (b), most candidates understood what was expected, with many getting both solutions correct. Others were unsure whether or not there was a second solution, and often did not give one or gave more than one. A common wrong answer was 256.7° , from $360^\circ - 103.3^\circ$.

Although some candidates answered part (c) confidently, many did not and some did not attempt it. There was confusion between maximum and minimum, and the common errors were to say that the minimum occurred at 90° or 180° with the expression equated to 0 or

–1 and not –5. Some thought the minimum value was an angle. The few attempts to use calculus were usually unsuccessful with candidates again unable to demonstrate that they had found a minimum value.

Q11.

Most candidates were able to gain some marks in part (a). In (i) the common wrong answer was a stretch in the x -direction with scale factor $1/2$. In (ii) most gave a correct reflection and in (iii) many gave the correct vector or its equivalent but still some candidates are not using the correct word ‘translation’. As previously stated, ‘tr’ was not accepted as a replacement for ‘translation’.

Part (b) was not answered well. The usual common error, illustrated by ‘ $\theta - 30 = 44.4$, $\theta = 74.4$ or $\theta = 180 - 74.4$ ’ was seen too frequently. It was not uncommon to see candidates giving answers in all four quadrants. A more surprising error from the better candidates is illustrated by ‘ $\theta - 30 = 44.4$, 135.6 so $\theta = 14.4$, 105.6 ’

Part (c) was answered well but the usual errors $(\cos x + \sin x)^2 = \cos^2 x + \sin^2 x$ and $(\cos x + \sin x)^2 = \cos^2 x + \sin^2 x + \cos x \sin x$ and the ‘trig identity ‘ $\cos x + \sin x = 1$ ’ were all seen.

Q12.

- (a) Candidates appeared to know the correct trigonometrical identity and this part was well answered.
- (b) This was also well answered with candidates able to factorise the expression from part (a).
- (c) This was reasonably well answered with most candidates obtaining two correct results and many fully correct answers seen. It was pleasing to see that there were very few candidates who used degrees. They were penalised.

Q13.

Parts (a) and (b) were done well and there were few errors, although some did make a sign error in finding angle α but didn’t respond to the given answer, and just made a comment about it being an acute angle ‘so is positive’.

Part (c) was found difficult by many candidates. Although most realised they were to use the results of part (a) rather than calculus, they confused the cosine having a maximum of 1 with the maximum value of the expression. Some candidates who had shown the maximum occurs at -33.7° , then subtracted from 90° or 180° to get a positive angle.

Q14.

This question was answered badly by the majority of candidates. In part (a) the examiners required the word ‘Stretch’ and either the correct direction or the correct scale factor before awarding either of the two marks. It was not uncommon to see the wrong scale

factor $\frac{1}{2}$ and the wrong direction (parallel to the y -axis). Many weaker candidates who attempted part (b) started by writing the incorrect statement ‘ $x = \tan^{-1}6$ ’. Many better

solutions were spoiled by manipulation errors illustrated by

$\frac{1}{2}x = 1.249, 4.3906. \Rightarrow x = 0.6245, 2.1958$. Part (c) was by far the worst answered part on the paper. Only a very few candidates realised that $\cos 9 = 0$ satisfied the given factorised form of the equation. The vast majority multiplied out the brackets and then cancelled the $\cos \theta$ factor. A mark was awarded to a significant number of candidates

for knowledge of the identity ' $\tan \theta = \frac{\sin \theta}{\cos \theta}$ ' in this part of the question.

Q15.

Part (a) Those candidates who appreciated $\sec x = 1/\cos x$ were usually able to evaluate the first answer of 1.37 radians. When candidates went on to complete this part correctly although 4.51 was a common error from $x + 1.37$. Answers in degrees were not common but they were seen.

Part (b) This part was answered very well by the majority of candidates when the most appropriate trigonometrical identity was used when candidates changed the equation into one involving $\sin x$ and $\cos x$, errors often did occur.

Part (c) Most candidates were able to successfully factorise the quadratic and many went on to complete the question correctly. Marks were lost by 'extra' values within the range being given. Candidates were not penalised for extra answers outside the given range.

Q16.

This question in general was not done well. A lot of candidates indicated a poor knowledge of trigonometric identities and a lack of experience in manipulating them. The Specification states that the double angle formulae should be learnt.

Part (a) Although most candidates could recollect or work out the identity for $\sin 2x$ this was not the case for $\cos 2x$ where many variations on incorrect answers were seen; some were correct but not in the requested form.

Part (b) Candidates were given credit for attempting to use their versions of $\sin 2x$ and

$\cos 2x$ together with $\tan x = \frac{\sin x}{\cos x}$. Those who started on the left hand side often omitted a common denominator of $\cos x$, some just putting it back into their work to get to the given answer. Those who started on the right hand side were generally more successful, but most candidates worked from both sides together, often making errors in their manipulations, such as omitting brackets and thus not multiplying out correctly. Many managed to make the two sides become equal after some invalid manipulation.

Part (c) Most candidates made the solution of this equation far more involved than was intended or is necessary. Few realised they could use the answer from part (b) and make use of, if $pq = 0$ then $p = 0$ or $q = 0$ and thus write down the solutions, with little more work. Some candidates did write down some correct solutions with no working seen, which was accepted. Many manipulated either form of the equation, into sines and cosines often making errors and abandoning the attempt. Many derived the cubic equation $\sin x - 2\sin^3 x = 0$ or something close to it, but still did not realise that $\sin x = 0$ was a solution. Of those who did get to correct solutions, the solution 225° , was often

missing.

Q17.

In part (a), the question asked for the coordinates of the marked points on the trigonometrical graph. For each point, examiners expected to be given both coordinates rather than just a non-zero value. Other common errors included answers involving radians and also answers which had the x - and y -coordinates switched. In part (b), candidates were expected to use the word “stretch” and also to give the direction and scale factor of the stretch. As expected, a scale factor of 2 rather than $1/2$ was the usual error.

Relatively few candidates gained all the available marks in part (c). Most obtained two

of the four values, but many missed the value from ‘ $x = 180 + 180 - \frac{\alpha}{2}$ ’. Candidates should also be aware that when asked to give answers to the nearest 0.1° , their earlier working should be done to a greater degree of accuracy. Failure to do this frequently lead to the wrong answer “ 214.2° ”.

Q18.

Most candidates were able to state and use the identity ‘ $\tan \theta = \frac{\sin \theta}{\cos \theta}$ ’ to obtain the printed result in part (a)(i). The identity ‘ $\cos^2 \theta + \sin^2 \theta = 1$ ’, required in part (ii), was less well recalled, with sign errors and ‘ $\cos \theta + \sin \theta = 1$ ’ seen. Many candidates correctly solved the equation in part (b)(i) but the explanations given for part (b)(ii) were poor. The most common wrong answer was ‘because $\cos \theta = -4$ is negative.’ Those who realised the link between the parts normally gained the marks for (b)(iii) and part (c), but many tried to solve part (c) from scratch, despite the clear instruction to ‘Write down’ the values.

Q19.

Candidates should know the trigonometrical identities as outlined in the specification. Part (a) was well answered and there were many correct solutions to part (b), although some scripts did not give fully convincing arguments. Part (c) required answers correct to the nearest degree, but answers to a greater degree of accuracy were not penalised. There were a significant number of candidates who obtained an answer of 109.47 which was rounded to 109.5 which was then rounded to 110.

Q20.

There was a very mixed response to this question with relatively few candidates gaining full marks. Some didn’t know how to start it at all.

Part (a) Most candidates found R successfully but made errors in finding the value of α with some finding α to be 63.4° . Most candidates responded to the request for α to be to one decimal place, though this created problems for some in part (b).

Part (b) Many candidates gave answers close to, but not exactly, zero, not realising they were rounding off the same angle to different accuracies. Some candidates did not



progress beyond $\sin(x + \alpha) = \sqrt{\frac{1}{5}}$ and, of those who did, many had problems in finding the second solution.